SC4001 NEURAL NETWORKS AND DEEP LEARNING · GROUP PROJECT · TOPIC E

Does Visual Contrastive Decoding Still Help When the Image Is Already Degraded?

Training-free hallucination mitigation for LLaVA-1.5 under blur and noise, and a view × layer readout that improves object grounding

Group [group number]

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Code: https://github.com/AkshayNarayananB/SC4001-NN (also submitted as a ZIP file)
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Does Visual Contrastive Decoding Still Help on Degraded Images?

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Abstract

Visual Contrastive Decoding (VCD) reduces object hallucination in vision–language models by contrasting the real image with a noised copy, implicitly assuming that the real image is clean. We test that assumption on LLaVA-1.5-7B and POPE under four levels each of Gaussian blur and noise. VCD's F1 gain does not shrink with degradation; it grows from +1.5 to +18.9 points. However, a baseline whose Yes/No threshold is merely shifted to VCD's yes-ratio matches it, and once forced answers are ranked fairly VCD's AUROC is unchanged: on degraded images VCD recalibrates rather than grounds. Under an image-level DEV/TEST protocol with threshold-matched baselines, stronger "blind" negatives, content-free null images, RITUAL and DeCo also leave discrimination essentially unchanged ($|\Delta\text{AUROC}| \leq 0.003$). We propose ViewLayer, a training-free readout that averages the Yes–No margin over lightly blurred and flipped views and reads it out at decoder layer 20, with all choices fixed on DEV images. On held-out images it raises AUROC in 26 of 27 condition $\times$ split cells (by +0.080 at the strongest noise) and F1 by up to 5.1 points over an equally tuned baseline; an ablation shows that the layer readout fixes blur and the views fix noise.

1 Introduction

Large vision–language models (VLMs) such as LLaVA-1.5 [1] often assert objects that are not in the image, a failure known as *object hallucination* [2, 3]. Visual Contrastive Decoding (VCD) [4] is a popular training-free remedy: at every decoding step it contrasts the logits obtained with the real image v against those obtained with a heavily noised copy v', boosting tokens whose score depends on actually seeing the image. VCD implicitly assumes that v is clean. Real inputs are often blurred or noisy; then the real branch already leans on language priors, the two branches should converge, and the contrast should carry little information—precisely when hallucination is most likely.

We therefore ask: *how do VCD and related training-free decoding methods behave when the input is already degraded, and can a training-free modification genuinely improve object grounding under blur and noise, rather than merely shift the model's Yes/No bias?* We study LLaVA-1.5-7B on the POPE benchmark [3] under nine conditions and make five contributions:

1. A faithful reproduction of VCD on clean POPE (within 0.5 F1 of the paper on every split), showing that the published numbers need diffusion step T=999 rather than the code default of 500 (Section 4.1).
2. A controlled degradation study showing that VCD's F1 gain *grows* with severity but is a recalibration toward "yes" that a threshold shift reproduces (Section 4.2).
3. An evaluation protocol that separates discrimination from bias: an image-level DEV/TEST split, threshold-matched baselines, and a decision-preserving AUROC that removes an artefact which had exaggerated VCD's discrimination loss roughly tenfold (Section 3.3).
4. A comparison of seven further modifications—blinder negatives, content-free nulls, restoration, RITUAL [5], DeCo [6] and entropy-routed layer readouts—with failure analyses explaining why they do not help (Sections 4.3 and 4.5).
5. ViewLayer, a view $\times$ layer readout that improves held-out AUROC in 26 of 27 cells and F1 by up to 5.1 points over an equally tuned baseline, with an ablation that attributes the gain on blur and on noise to different components (Sections 3.5 and 4.4).

2 Review of existing techniques

Measuring hallucination. CHAIR [2] counts objects in generated captions that are absent from the annotations. POPE [3] asks balanced yes/no questions ("Is there a dog in the image?") about MSCOCO [7] images, sampling absent objects at random, by frequency (*popular*) or by co-occurrence (*adversarial*). Because POPE is binary, its F1 depends strongly on how often a model says "yes", so the yes-ratio is reported alongside it.

Contrastive decoding. VCD [4] subtracts the logits of a diffusion-noised image [8] from those of the real image and keeps only tokens that are plausible under the real image. Visual Debias Decoding [9] contrasts against a meaningless image instead. Both estimate the language prior from a corrupted or absent visual input and remove it.

Attention and internal states. OPERA [10] penalises over-trust patterns during beam search, and PAI [11] amplifies attention to image tokens. For text-only language models, DoLa [12] contrasts the final layer with a

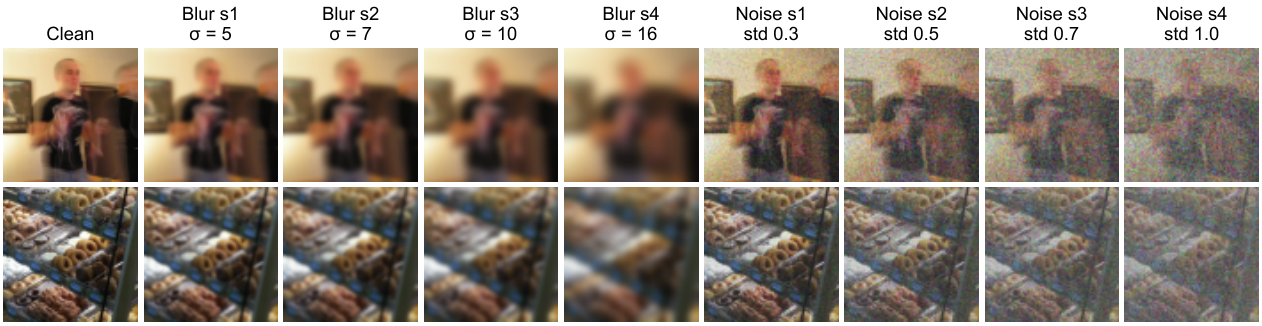


Figure 1: Two POPE images under the nine conditions, shown at the model's 336 px input size. Blur is parameterised by σ (px) and noise by its standard deviation on a $[0, 1]$ intensity scale.

"premature" layer to improve factuality. DeCo [6] observes that multimodal models often recognise an object in their middle layers even when the final layer denies it, and mixes the most confident preceding layer back into the final logits.

Multiple views and robustness. RITUAL [5] *adds* the probabilities obtained from a randomly transformed copy of the image, akin to test-time augmentation [13]. Separately, common corruptions such as blur and noise are known to degrade vision models sharply [14], and restoring the input, for example with a denoising autoencoder [15], is a standard counter-measure.

Gap. All of these decoding methods were developed on clean COCO images and are usually compared on F1 alone. None studies degraded inputs, and none separates a gain in discrimination from a shift in Yes/No bias.

3 Methods

3.1 Model, benchmark and degradations

We use LLaVA-1.5-7B [1] (`llava-hf/llava-1.5-7b-hf`; a CLIP ViT-L/336 encoder [16] feeding 576 visual tokens into a 32-layer Vicuna-7B decoder) in fp16 with HuggingFace Transformers 4.46.3 [17], on one A100-40GB of NSCC ASPIRE 2A. POPE-MSCOCO has three splits (random, popular, adversarial) of 3,000 balanced questions over the same 500 val2014 images. The prompt follows the official VCD code (question + "Please answer this question with one word.").

Degradations are applied at native resolution, before LLaVA resizes the image to 336 px (Figure 1): Gaussian blur with $\sigma \in \{5, 7, 10, 16\}$ px and additive Gaussian noise with std $\in \{0.3, 0.5, 0.7, 1.0\}$ on a $[0, 1]$ scale, clipped and seeded per image. The levels were calibrated in a 100-question pilot so that baseline accuracy falls steadily (by about 3/7/14/23 points for blur and 2/12/14/20 for noise); a first, weaker noise ladder (std ≤ 0.2) barely affected the model because the resize averages out pixel noise.

3.2 Baseline and VCD

Decoding is greedy, so POPE's one-word answer is fixed by the first generated token. We summarise each question by its first-token margin $m = \ell(\text{Yes}) - \ell(\text{No})$; the baseline answers "yes" iff $m > 0$. VCD replaces the logits by

$$s(y) = (1 + \alpha)\,\ell(y \mid v, x) - \alpha\,\ell(y \mid v', x), \tag{1}$$

and masks every token whose probability under v is below β times the top probability. Here v' is v after forward diffusion to step T [8]; we use α=1, β=0.1, T=999. The mask can *force* the answer: if "No" is implausible, VCD answers "Yes" whatever the contrast says.

3.3 Evaluation protocol

All methods are scored by one shared implementation (`fix_eval/common.py`).

DEV/TEST split. The 500 images are split once (seed 0) into 100 DEV and 400 TEST images, giving 2,400 TEST questions per condition $\times$ split cell. Every hyperparameter (α, views, layer, threshold) is chosen on DEV by its mean over all 27 cells, so each method has one global setting and is never told how degraded an image is. All reported numbers are on TEST.

Discrimination versus bias. AUROC of a method's first-token score measures how well it ranks present above absent objects, independent of the threshold. The *matched-threshold baseline* thresholds the base-

Table 1: Modifications investigated. "Passes" counts image forward passes per question (baseline: 1).

Idea	Method (owner)	What changes	Passes
Blinder negative	Zero-hook VCD (Akshay)	Contrast branch receives 576 zeroed visual tokens	2
	Shuffle-hook VCD (Akshay)	Contrast branch receives randomly permuted patch embeddings	2
Content-free null	Phase null (Sanjana)	Contrast image keeps the amplitude spectrum, randomises phase [19]	2
	Mean null (Sanjana)	Control: contrast image is the per-channel mean	2
Restoration	Fix B (**[owner]**)	**[DAE trained on (degraded, clean) COCO pairs restores the input before VCD]**	**[?]**
Extra views	RITUAL [5] (Sanjana)	$p(y\|v)+\alpha\, p(y\|T(v))$ with a random transform T; also six views	2 (7)
Layer readout	DeCo [6] (Sanjana)	Final logits + $\alpha\, p_{\max}\cdot$ logits of the most confident layer in 20–28	1
	Entropy routing (Akshay)	First token from layers $\{18,\dots,30\}$ weighted by softmax$(-H/0.5)$	1
Views $\times$ layer	**ViewLayer (Sanjana)**	Mean Yes–No margin over views at a DEV-chosen layer (Section 3.5)	5

line margin so that its yes-ratio equals the method's: it is what a pure bias shift achieves, and a genuine improvement must beat it in F1.

Decision-preserving AUROC. Masked methods output $\pm\infty$ for forced answers—60–70% of VCD's answers are forced—and ranking these as ties penalises the method arbitrarily. We rank forced answers among themselves by the real-branch margin, which keeps the ranking consistent with the decisions actually made, and also report the unmasked score.

Uncertainty. We use 95% paired bootstrap intervals over questions (500 resamples) [18] and count the cells whose interval lies entirely above or below zero.

3.4 Modifications investigated

Table 1 lists the modifications the team investigated, grouped by the idea behind each. Stage 2 (Section 4.2) motivated the first two groups: on clean images VCD's "blind" image still separates present from absent objects with AUROC 0.69, so subtracting it also subtracts evidence. The zero and shuffle hooks remove visual content inside the network; the phase-scrambled null removes it in pixel space while keeping the input's amplitude spectrum, so a blurred input yields a blurry, object-free texture (Figure A.1). RITUAL was re-implemented from the paper and DeCo ported from its official `deco_greedy_search`, verified by a CPU test against a line-by-line copy of the official logic. In the layer-routing experiments, a DoLa-style router that picks the layer of maximum Jensen–Shannon divergence from the final layer [12] was tried first and then replaced by entropy-weighted soft routing of the first token.

3.5 ViewLayer: a view $\times$ layer readout

ViewLayer combines the two signals that our Phase 2 runs found to survive degradation (Section 4.3): in the RITUAL ablation a slightly blurred copy of a noisy image raised AUROC, acting as a free denoiser, and in the DeCo layer analysis the margin read out at layer $\approx$20 separated present from absent objects better than the final layer, increasingly so with severity. DeCo's mixing weight $\alpha p_{\max}$ is too small to exploit the latter, and RITUAL uses only the final layer.

For a view set $V \subseteq \{\text{original, blur } \sigma{=}1,2,3 \text{ px, horizontal flip}\}$, applied to the degraded image at native resolution, and a decoder layer L, the score is the mean Yes–No margin of layer L read out through the final RMSNorm and LM head, as in DoLa and DeCo (Figure 2):

$$s(q) = \frac{1}{|V|}\sum_{u\in V}(w_{\text{Yes}} - w_{\text{No}})^{\top}\,\text{RMSNorm}(h_L(u,q)), \qquad \hat{y} = \text{“yes”} \iff s(q) > t, \tag{2}$$

where h_L is the hidden state at the last prompt position and w are rows of the LM head. V and L maximise mean DEV AUROC over nine candidate view sets and layers 12–32; t maximises mean DEV F1. Because t is tuned, the baseline also receives its own DEV-tuned threshold. All views share one batched forward pass.

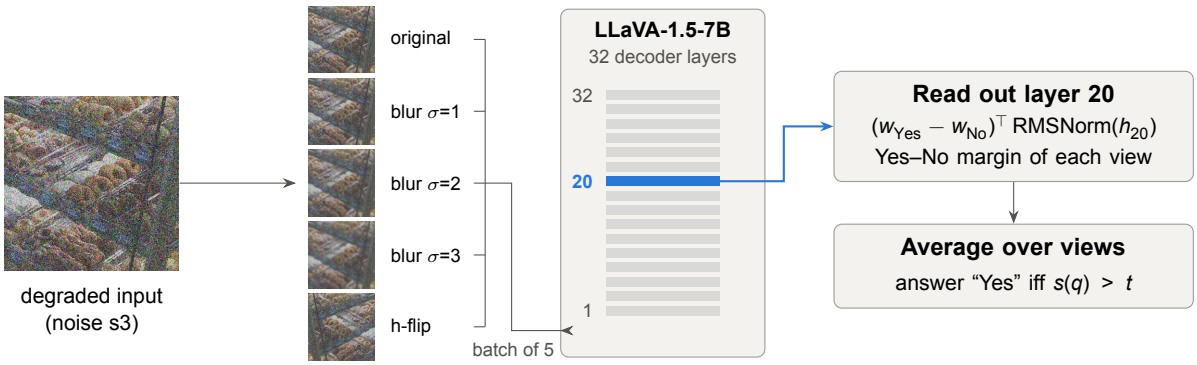


Figure 2: ViewLayer. The degraded image and four light variants are encoded in one batched forward pass. Instead of the final layer, the Yes–No margin is read out at decoder layer 20 for every view and averaged. Views, layer and threshold are chosen on DEV images only.

Two ablations are selected the same way: *layer-only* (original image, best layer) and *views-only* (best views, final layer).

Disclosure. The idea of a blur view came from the RITUAL ablation, which we had inspected on TEST images. Every *choice*—views, layer and threshold—was made on DEV images only, but the candidate list was informed by those earlier results.

4 Experiments and results

4.1 Reproducing VCD on clean images

Before testing degraded inputs we verified our implementation: the noise function is bit-identical to the official code for the same random stream, and the metric code matches theirs exactly. With T=999, VCD lands within 0.5 F1 of the paper on every split and the baseline within 0.25 (Table 2); greedy results on the random split match the paper's appendix to within 0.1 on every metric. At the code default T=500 the gain roughly halves, which explains several open reproduction issues on the official repository; all later experiments use T=999. Even on clean images most of VCD's gain comes from saying "yes" more often (yes-ratio 39–44% $\rightarrow$ 46–52%).

Table 2: POPE F1 (%) on clean images with sampling (temperature 1); mean $\pm$ std over three seeds, 3,000 questions per split.

Split	Baseline (ours)	Baseline (paper)	VCD, T=500 (ours)	VCD, T=999 (ours)	VCD (paper)
Random	81.19 $\pm$ 0.45	81.33	83.53 $\pm$ 0.30	**87.09** $\pm$ 0.25	87.16
Popular	80.20 $\pm$ 0.78	80.06	82.48 $\pm$ 0.81	**85.48** $\pm$ 0.33	85.06
Adversarial	77.81 $\pm$ 0.29	77.57	79.62 $\pm$ 0.27	**81.77** $\pm$ 0.31	81.33

4.2 VCD on degraded images

Table 3 compares the greedy baseline and VCD on TEST images.

VCD's F1 gain grows with severity. It rises from +1.5 on clean images to +18.6 at blur s4 and +18.9 at noise s4 (over all 500 images: +1.2, +19.5, +20.1, every degraded cell significant). Measured by F1, the hypothesis that VCD stops helping on degraded images is refuted.

But the gain is a threshold shift. Degradation pushes LLaVA toward "no": its yes-ratio falls from 41% to 16–17% at s4 while precision stays near 87–94% and recall collapses. VCD pushes back toward "yes", and the matched-threshold baseline equals or beats VCD's F1 in every condition. The team's Stage 2 analysis had found VCD's AUROC below the baseline's in all 27 cells by 0.016–0.054; re-scored with the decision-preserving rule, that drop is almost entirely a tie artefact. Ranked fairly, VCD is within 0.005 of the baseline everywhere (mean -0.0024), and its unmasked score is slightly *better* at s4.

The predicted mechanism is nevertheless real. The Jensen–Shannon divergence between the branches halves under degradation (0.135 $\rightarrow$ 0.066 at blur s4), and the discriminative part of the contrast—its mean on present minus absent objects—falls from 3.79 to 1.83. For absent objects the contrast even turns positive ($-0.38 \rightarrow$ +1.07 at noise s4). The contrast loses image information as predicted, but its large constant push

Table 3: Greedy baseline vs VCD (α=1, β=0.1, T=999) on TEST images, mean of three splits. *Matched*: baseline thresholded to VCD's yes-ratio. †Original Stage 2 analysis (all 500 images, forced answers ranked as ties).

	F1 (%)			Yes-ratio (%)		AUROC			
Condition	Base	VCD	Matched	Base	VCD	Base	VCD, ties†	VCD, dec.-pres.	VCD, unmasked
Clean	84.2	85.7	**86.8**	40.8	49.2	.940	.921	.935	.925
Blur s1	75.9	82.3	**83.0**	34.3	47.4	.907	.884	.904	.897
Blur s2	70.6	79.0	**79.4**	31.0	45.3	.877	.855	.875	.871
Blur s3	61.0	73.1	**73.4**	25.2	40.6	.836	.800	.834	.833
Blur s4	41.1	59.7	59.7	15.5	30.9	.750	.708	.749	.756
Noise s1	76.4	81.9	**82.6**	34.9	46.6	.910	.889	.906	.899
Noise s2	69.7	78.0	**78.7**	30.3	45.9	.878	.851	.874	.871
Noise s3	58.4	72.5	**72.6**	23.7	43.3	.823	.796	.821	.827
Noise s4	43.8	62.7	**62.8**	17.0	38.2	.737	.708	.737	.748

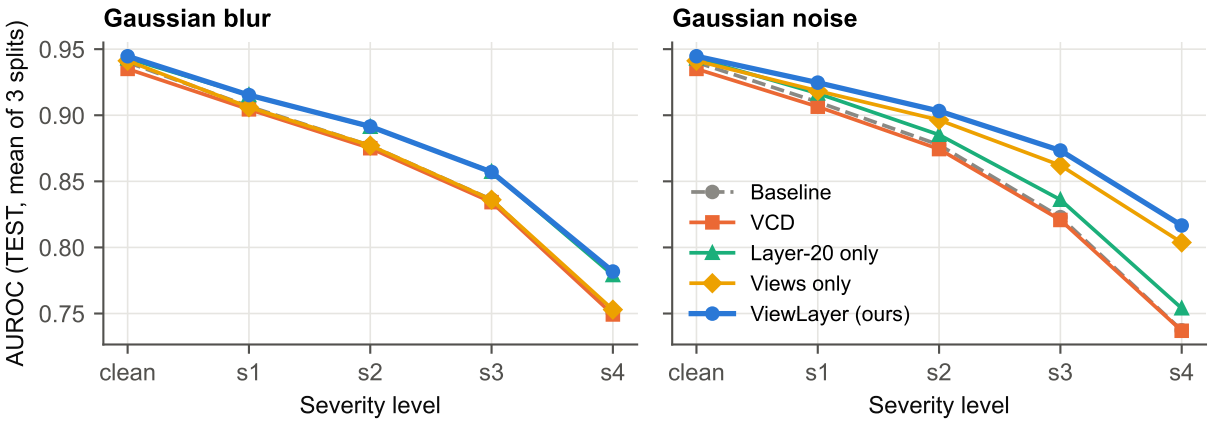


Figure 3: TEST AUROC against severity (mean of three splits). The layer-20 readout accounts for the gain on blur and the blurred views for the gain on noise; ViewLayer combines both.

toward "yes" happens to correct the degradation-induced bias toward "no".

4.3 Comparison of all modifications

Table 4 scores every method on the same TEST images. Every method that keeps the final-layer readout of the original image barely changes discrimination: mean ΔAUROC stays within $\pm$0.003, F1 rarely beats the matched-threshold baseline, and F1 differences between methods mostly track their yes-ratios. The phase null is significantly *worse* than the baseline in 17 cells; RITUAL as published lowers F1 under degradation (blur s4: 41.1 $\rightarrow$ 31.2); its six-view variant gives the most consistent final-layer AUROC gain (15 cells) but an even lower F1. Entropy routing matches the baseline on clean images and is within +0.004 at s4. Only readouts that leave the final layer, the original view, or both, give large and consistent gains.

4.4 ViewLayer results and ablation

On DEV the selected setting was all five views read out at **layer 20** (mean DEV AUROC 0.879 vs 0.857 for the baseline); layer 19 was a close second, and the best single-view layer was also 20 (Figure 4b). On TEST (Figure 3), ViewLayer's AUROC is significantly above the baseline's in 26 of 27 cells—the exception, clean/adversarial, has an interval touching zero—and the gain grows with severity: +0.005 on clean images, +0.032 at blur s4 and +0.080 at noise s4 (95% intervals +0.062 to +0.094 across splits).

Ablation. The two components address different degradations. Layer 20 alone provides nearly all of the gain on blur (blur s4: .779 vs .782) but little on noise; the views alone provide most of the gain on noise (noise s4: .804) and nothing on blur (.753), since blurring an already blurred image adds no information. Their mean gains (+0.013 and +0.015) add up to ViewLayer's +0.028. Figure 4a shows why the layer helps: the per-layer AUROC peaks around layers 19–20, and the advantage over layer 32 grows from +0.003 on clean images to +0.029 at blur s4.

Table 4: All methods on TEST images (27 condition × split cells). Sig. ↑/↓: cells whose 95% bootstrap interval lies above/below zero, against the baseline (AUROC) or the matched-threshold baseline (F1). F1 of the rows marked ⋆ uses DEV-tuned thresholds and is directly comparable among them; other rows use their own decision rule. ‡Reported on all 500 images (baseline .938/.754/.736).

		Mean	Sig. ↑/↓		AUROC			F1 (%)	
Method	Passes	ΔAUROC	AUROC	F1	Clean	Blur s4	Noise s4	Blur s4	Noise s4
Baseline (threshold 0)	1	–	–	–	.940	.750	.737	41.1	43.8
Baseline (DEV threshold)⋆	1	–	–	–	.940	.750	.737	67.9	68.8
VCD	2	−.0024	–	–	.935	.749	.737	59.7	62.7
Zero / shuffle-hook VCD	2	[?]	[?]	[?]	[?]	[?]	[?]	[?]	[?]
Phase null (α=.25)	2	−.0008	2/17	1/1	.939	.749	.737	47.3	51.2
Mean null (α=.25)	2	+.0003	11/0	1/0	.940	.751	.738	50.2	53.7
Fix B: DAE + VCD	[?]	[?]	[?]	[?]	[?]	[?]	[?]	[?]	[?]
RITUAL (paper, α=3)	2	−.0022	0/6	0/5	.939	.750	.735	31.2	35.0
RITUAL (DEV α=0.5)	2	+.0006	2/0	3/0	.941	.751	.739	37.3	40.2
RITUAL, 6 views (DEV α=5)	7	+.0025	15/0	4/0	.942	.751	.743	28.8	30.7
DeCo (paper, α=0.6)	1	+.0008	13/1	0/1	.940	.752	.738	50.9	54.1
DeCo (DEV α=1.5)	1	+.0008	12/1	5/1	.940	.752	.738	52.9	56.3
Entropy routing‡	1	n/a	n/a	n/a	.938	.758	.738	[?]	[?]
Layer 20 only⋆	1	+.0133	25/0	3/1	.943	.779	.754	67.5	67.7
Views only⋆	5	+.0151	15/0	11/0	.941	.753	.804	67.9	73.6
ViewLayer⋆	5	**+.0277**	**26/0**	**10/0**	**.945**	**.782**	**.817**	**68.6**	**73.9**

Table 5: Operating points on TEST images (%, mean of three splits). Precision, recall and false-positive rate (FPR) follow exactly from F1 and the yes-ratio because every cell is balanced. Full results: Appendix B.

	Blur s4					Noise s4				
Method	Yes	Prec.	Rec.	FPR	F1	Yes	Prec.	Rec.	FPR	F1
Baseline (threshold 0)	15.5	87.4	26.9	4.0	41.1	17.0	86.8	29.3	4.6	43.8
Baseline (DEV threshold)	43.1	73.6	63.1	23.1	67.9	53.4	66.8	71.0	35.7	68.8
VCD	30.9	78.7	48.2	13.5	59.7	38.2	73.0	55.2	21.2	62.7
ViewLayer	43.6	**74.0**	**64.1**	23.1	**68.6**	56.3	**70.1**	**78.3**	34.2	**73.9**

Operating points. Table 5 decomposes F1 into precision, recall and false-positive rate (absent objects answered "yes", i.e. hallucinations). Against the baseline with an equally tuned threshold, ViewLayer improves F1 by +1.3 points on average and by +3.4/ + 5.1 at noise s3/s4, where it is better on precision, recall *and* false-positive rate simultaneously—so the gain is not a bias shift. It beats the threshold-matched baseline significantly in 10 cells and is never significantly worse.

4.5 Failure analysis

VCD adds hallucinations along with recall. Under degradation the contrast is positive for absent objects, so VCD's false-positive rate rises from 4.6% to 21.2% at noise s4 while its AUROC is unchanged: its F1 gain is paid for with hallucinations that a threshold change would produce equally.

RITUAL amplifies the "no" bias. Its views are degraded too and lean "no" as much as the original, so adding α=3 times their probability strengthens the bias (yes-ratio at blur s4: 15.5% → 10.6%) and F1 drops. Individually, vertical flip, rotation, colour jitter and crop lower AUROC in most conditions; only horizontal flip (neutral) and the Gaussian-blur view, which helps noisy images (.737 → .751 at noise s4), do not (Appendix C). This observation motivated ViewLayer's blur views.

Null images are not content-free. On its own, the phase-null branch still ranks present above absent objects with AUROC 0.56–0.58 on blurred inputs: the amplitude spectrum carries scene gist [20] (a savanna texture raises the prior for "giraffe"), so subtracting it removes evidence. The mean null has AUROC 0.43, below chance, so subtracting it adds a little signal—hence its 11 small gains.

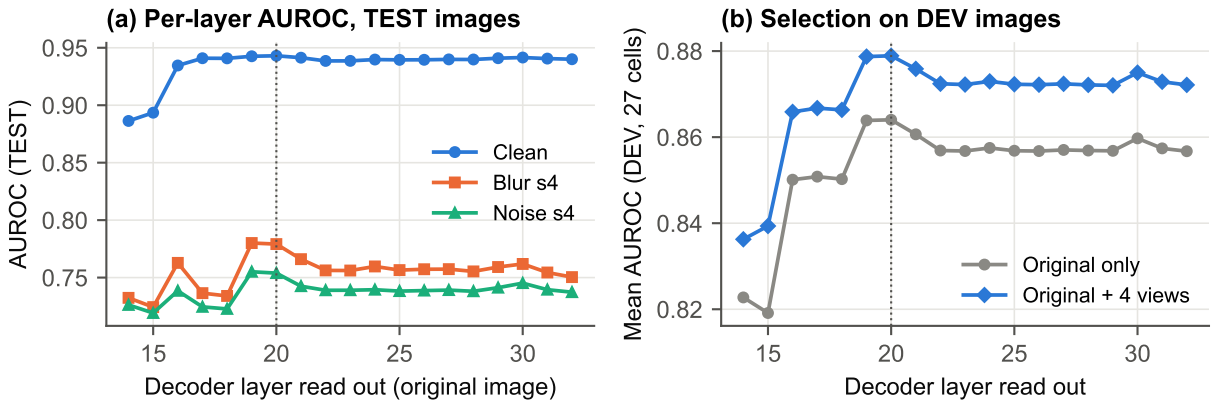


Figure 4: (a) TEST AUROC of the Yes–No margin read out at each decoder layer from the original image. (b) DEV selection: mean DEV AUROC over all 27 cells against layer, with and without the four added views.

Divergence-based layer routing collapses. On degraded inputs the final-layer distribution is too flat for the JSD criterion to be informative; the router chose the earliest candidate layer, which produced malformed text without an end-of-sequence token and no parsable answers. Entropy weighting removed the collapse but only matched the baseline.

Where ViewLayer fails. (i) *Clean images*: AUROC rises by only +0.005 and F1 is 0.2 below the DEV-threshold baseline. (ii) *Adversarial split*: gains are smallest (noise s4: +4.0 F1 vs +6.3 on random), because co-occurring absent objects carry the strongest prior. (iii) *Blur*: the views do not help, so the gain rests on layer 20 alone. (iv) *Operating point*: one F1-optimal global threshold gives yes-ratios of 44–62% and false-positive rates of 23–34% (baseline at threshold 0: 4–5%); the DEV-tuned baseline behaves the same, so this reflects F1 tuning on balanced data rather than the score, and a hallucination-averse deployment should use a stricter threshold. (v) *Lost information*: even at .817, noise s4 AUROC stays far below clean (.945); views and layers recover evidence the model computed but ignored, not evidence the degradation destroyed.

5 Discussion

What we learned about VCD. The original hypothesis was right about the mechanism and wrong about the outcome. Degradation does make the branches converge and drains the contrast of image information, but on POPE the dominant failure under degradation is *missing* objects, and VCD's steady push toward "yes" compensates for it. F1 alone cannot distinguish a better decoder from a recalibrated one; we recommend reporting AUROC with a decision-preserving tie rule and a threshold-matched baseline for any decoding-time method. Without the tie rule, our own Stage 2 analysis overstated VCD's discrimination loss roughly tenfold.

Why ViewLayer works. It exploits two sources of evidence that the final-layer readout of the original image discards. Averaging over lightly blurred views is test-time augmentation [13] that suppresses pixel noise before the encoder; the intermediate-layer readout recovers evidence that the final layers overwrite with language priors, as DeCo [6] reported on clean images—we find this effect small on clean images but growing with degradation. Because the two sources repair different degradations, their gains add. And since ViewLayer subtracts nothing, it sidesteps the failure shared by every contrastive variant we tried: there is no reliably blind image to subtract.

Novelty. RITUAL and DeCo were proposed and evaluated on clean images, using only the final layer and only the original view, respectively; to our knowledge neither has been studied on degraded inputs or combined. Our contributions are a controlled study of training-free decoding under graded blur and noise, an evaluation protocol that separates discrimination from bias, well-supported negative results for blinder negatives, content-free nulls and published multi-view decoding, and a simple combination whose components are shown by ablation to address different degradations.

Limitations. We test one model, one benchmark and two synthetic degradations; low light, compression and motion blur are untested. ViewLayer scores the first token and so applies to yes/no questions; free-form captioning (CHAIR [2]) would need a per-token readout. The DEV set is small (100 images), the blur-view candidate was suggested by an earlier TEST observation, and the bootstrap resamples questions, which are correlated within an image, so intervals are somewhat optimistic. Five batched views cost 0.33 s per question on an A100, against 0.08 s for the baseline and 0.16 s for VCD.

Future work. A per-token ViewLayer for free-form generation evaluated with CHAIR; other VLMs and real-world corruptions; and views chosen adaptively from an estimate of the noise level.

6 Conclusion

On degraded images, VCD's large F1 gains on POPE come from recalibrating LLaVA's degradation-induced reluctance to say "yes", not from better visual grounding: a threshold change achieves the same, and once forced answers are ranked fairly its discrimination is unchanged. Stronger negatives, content-free nulls, RITUAL and DeCo do not change this picture. Reading the model out at layer 20 and averaging over lightly blurred views does: with every choice fixed on DEV images, ViewLayer improves held-out AUROC in 26 of 27 cells and F1 by up to 5 points over an equally tuned baseline, with the largest gains under heavy noise—the setting this project set out to fix.

AI Use Statement

We used **Claude Code** (Anthropic, Claude Opus model) as a coding and research assistant: to search the literature for training-free hallucination fixes; to implement and test the evaluation package (`fix_eval`), the phase/mean nulls, the RITUAL and DeCo re-implementations and ViewLayer; to submit and monitor ASPIRE jobs; to produce the analysis tables and figures; and to draft this report. We verified its outputs with 21 CPU unit tests (including a check of the DeCo port against a line-by-line copy of the official code), smoke runs before every full run, and sanity checks: the layer-32 readout reproduces the Stage 2 baseline margin (maximum difference 0.10 logits, from fp16 batching), and margins recomputed from logged logits agree with the decoded answers on 100% of questions. References were checked against arXiv and conference pages. Errors introduced by the assistant, such as rounding probabilities to six decimals, were caught and corrected. **[Arushi and Akshay: add the AI tools you used and how you checked them.]** The team chose the research question, degradations and evaluation criteria, decided which methods to pursue or drop, ran the experiments and interpreted the results; every number reported comes from experiments the team ran.

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A Reproducing the results

Dependencies. Python 3.11; PyTorch 2.5.1 (CUDA 12.1) and torchvision; pinned packages in `requirements.txt` (transformers 4.46.3, accelerate 1.1.1, pillow 11.0.0, numpy 1.26.4, scipy, scikit-image, pandas, pyyaml). On ASPIRE, `scripts/setup_aspire.sh` builds the conda environment. Weights are downloaded once on a login node (`huggingface-cli download llava-hf/llava-1.5-7b-hf`); jobs then run with `HF_HUB_OFFLINE=1`.

Data. Set `VCD_DATA_ROOT` and `VCD_OUTPUT_ROOT`, then run `python -m scripts.download_pope` (login node; POPE questions and 500 COCO images), `python -m scripts.make_degraded` (clean, blur s1–s4 and noise s1–s4 PNGs and `manifest.csv`) and `python -m scripts.contact_sheet` (Figure 1).

Baseline and VCD (Stages 0–2). `python -m scripts.make_configs stage2` writes the run configs; `bash scripts/submit.sh configs/stage2/` submits them; `python -m scripts.analyze --stage stage2` writes the comparison, diagnostic and margin tables.

Phase 2 methods. Each folder contains a PBS array job over the nine conditions, an analysis script that fills the folder's README and writes `results/<method>_*.csv`, and CPU tests:

Run (ASPIRE)	Analyse
`qsub model_fix_2/run_nulls.pbs`	`python -m model_fix_2.analyze`
`qsub RITUAL/run_ritual.pbs`	`python -m RITUAL.analyze`
`qsub DeCo/run_deco.pbs`	`python -m DeCo.analyze`
`qsub ViewLayer/run_viewlayer.pbs`	`python -m ViewLayer.analyze`

Adding `-J 4-8:4 -v LIMIT=30` to `qsub` runs a 30-question smoke test, and `python -m pytest ViewLayer/tests RITUAL/tests DeCo/tests model_fix_2/tests -q` runs the CPU tests. The zero/shuffle hooks and entropy routing (`vcd_zero`, `vcd_shuffle`, `dola_anchor` in `src/decode.py`, branch `akshay-branch`) run through `scripts/run_pope.py` and are analysed with `python -m scripts.analyse_dola --stage <stage>`. **[Fix B: add commands.]**

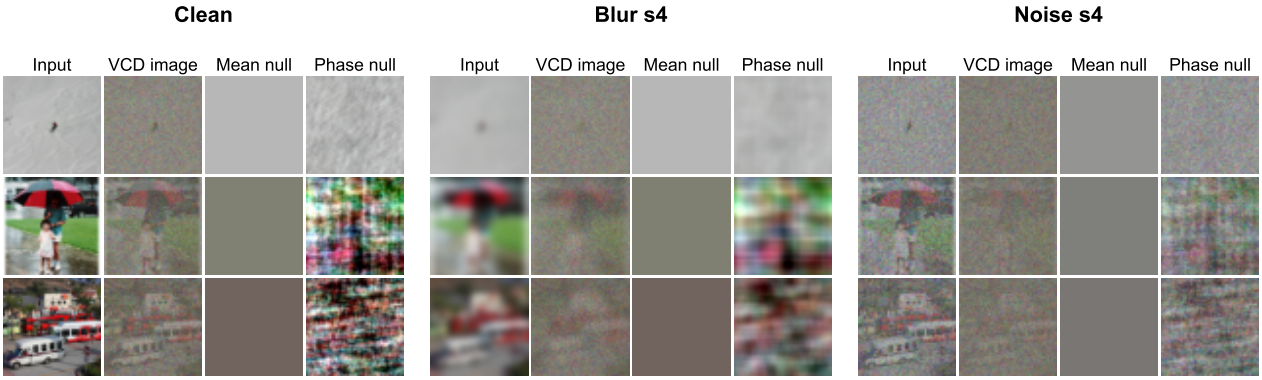


Figure A.1: Contrast images for three inputs. VCD's *T*=999 image still shows faint object outlines; the mean null is uniform; the phase null keeps the input's blur or graininess but no objects.

B Per-condition ViewLayer results

Table B.1: TEST images, mean of three splits. VCD AUROC uses the decision-preserving rule. Per-split values with bootstrap intervals: `results/viewlayer_comparison.csv`.

	AUROC			F1 (%)				
Condition	Base	VCD	ViewLayer	Base (thr. 0)	Base (DEV thr.)	VCD	ViewLayer	Yes% ViewLayer
Clean	.940	.935	**.945**	84.2	84.7	**85.7**	84.5	61.8
Blur s1	.907	.904	**.915**	75.9	83.2	82.3	**83.4**	57.9
Blur s2	.877	.875	**.892**	70.6	80.6	79.0	**80.8**	55.9
Blur s3	.836	.834	**.857**	61.0	77.3	73.1	**77.8**	52.6
Blur s4	.750	.749	**.782**	41.1	67.9	59.7	**68.6**	43.6
Noise s1	.910	.906	**.925**	76.4	82.9	81.9	**83.7**	59.6
Noise s2	.878	.874	**.903**	69.7	80.5	78.0	**81.7**	58.4
Noise s3	.823	.821	**.873**	58.4	75.6	72.5	**79.0**	57.4
Noise s4	.737	.737	**.817**	43.8	68.8	62.7	**73.9**	56.3

C RITUAL per-transform ablation

Table C.1: TEST AUROC of RITUAL with each single transform as the view (α=3), mean of three splits.

Condition	Baseline	H-flip	V-flip	Rotate	Colour jitter	Gaussian blur	Crop
Clean	.940	.941	.937	.938	.939	.939	.936
Blur s1	.907	.908	.900	.905	.904	.906	.904
Blur s2	.877	.878	.874	.875	.874	.876	.874
Blur s3	.836	.837	.830	.835	.830	.836	.833
Blur s4	.750	.751	.746	.750	.747	.750	.748
Noise s1	.910	.911	.906	.907	.908	**.912**	.907
Noise s2	.878	.881	.870	.868	.872	**.883**	.875
Noise s3	.823	.824	.816	.815	.817	**.836**	.818
Noise s4	.737	.739	.731	.729	.732	**.751**	.732